Submitted By:

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Assignment:

Pacman Game Report

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

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PACMAN CONSOLE GAME IN JAVA

PROJECT REPORT

1. INTRODUCTION

This project is a console-based Pacman game developed in Java using Object-Oriented Programming (OOP) principles. The game runs entirely in the terminal and does not use any external libraries. It is designed to demonstrate how multiple classes can interact with each other to create a complete working system that includes game logic, user interaction, and real-time updates.

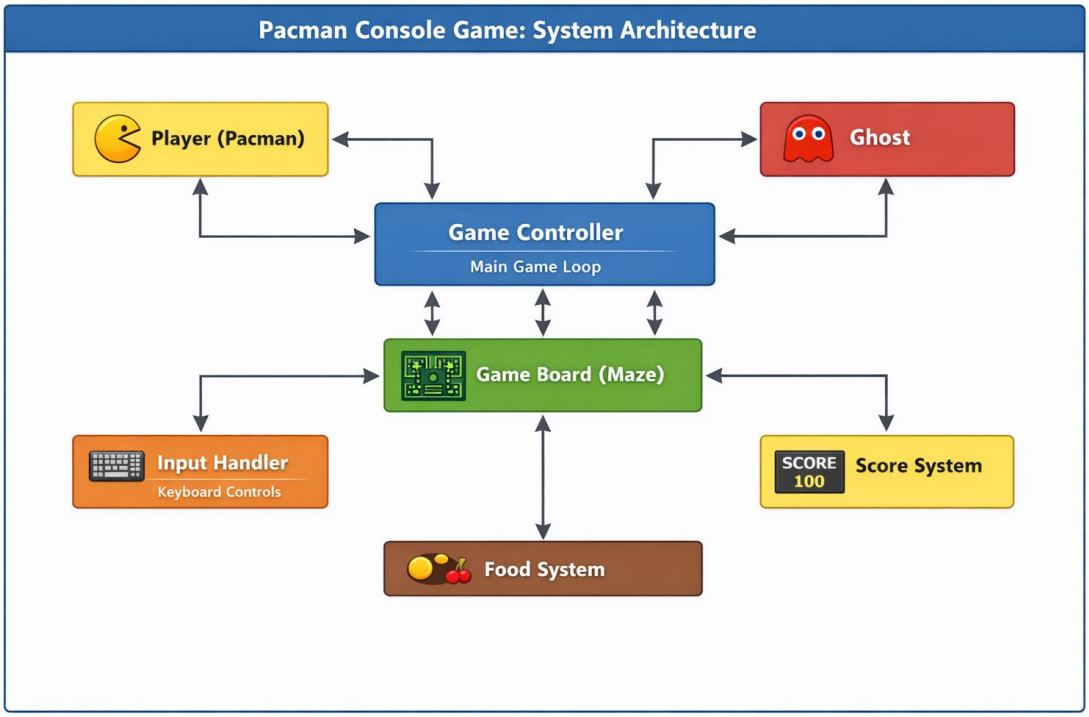
In this game, the player controls Pacman using keyboard inputs and navigates through a maze to collect food items while avoiding a ghost. The game continues until either all food is collected or Pacman collides with the ghost.

2. OBJECTIVE

The main objective of this project is to apply object-oriented programming concepts in a practical scenario by developing a simple interactive game. It aims to demonstrate how different classes can work together in a structured system to manage movement, scoring, rendering, and game state.

Another important goal is to understand how a continuous game loop works in a console-based environment, including handling user input and updating the game state after each move.

3. SYSTEM ARCHITECTURE



4. GAME FLOW

The game begins by initializing all required objects, including the game board, player, ghost, and food system. Once initialization is complete, the game enters a continuous loop that runs until a win or lose condition is met.

During each iteration of the loop, the game board is displayed in the console. The player is then prompted to enter a movement command, which is used to update Pacman's position. If Pacman moves to a location containing food, the food is collected and the score increases.

After the player's move, the ghost makes its move randomly within the maze. The system then checks for collisions between Pacman and the ghost. If a collision occurs, the game ends with a loss. If all food items are collected, the game ends with a win.

5. CONCLUSION

This project successfully demonstrates the use of Object-Oriented Programming in Java through the development of a simple console-based game. It shows how multiple classes

can be designed to work together in a structured architecture to manage different aspects of a system such as input handling, game logic, rendering, and state management.

The project also provides practical understanding of key programming concepts such as class interaction, encapsulation, modular design, and game loop implementation. It serves as a strong foundation for developing more advanced and complex game systems in the future.